

Quantum Vision Defect Detection System

Quant-A-thon 2026 | Task QT-2.3

Submission Report - July 09, 2026

1. Problem Statement

Manufacturing quality inspection requires fast, accurate detection of product defects. Manual inspection is slow and inconsistent. This project delivers an automated Quantum Vision-based Defect Detection System that analyzes product images and generates inspection reports with pass/fail verdicts.

2. Objectives

- Detect product defects in manufacturing images
- Reduce inspection time through automated analysis
- Improve quality assurance with confidence-scored reports

3. Solution Overview

We built a hybrid quantum-classical vision pipeline. Classical computer vision extracts edge density, texture variance, patch anomaly scores, and contour metrics. These features are angle-encoded into a 4-qubit variational quantum circuit (PennyLane). Classical and quantum predictions are fused to produce a final verdict with confidence.

4. Deliverables

- Image Analysis Model: Classical CV + 4-qubit VQC (PennyLane)
- Defect Detection Report: Downloadable TXT and JSON reports from the UI
- User Interface: Streamlit web application with visual heatmaps and metrics

5. Architecture

Upload Image -> Feature Extraction (OpenCV)

- > Classical Model (weighted scoring)
- > Quantum Vision Model (4-qubit VQC, angle encoding)
- > Hybrid Fusion (55% quantum / 45% classical)
- > Final Verdict + Detection Report

6. Quantum Innovation

- Angle encoding of 8 image features onto 4 qubits (RY + RZ rotations)
- 3-layer variational circuit with ring CNOT entanglement topology
- PauliZ measurement mapped to defect probability via sigmoid activation
- Captures non-linear feature correlations beyond classical linear models

7. Tech Stack

- OpenCV - image processing and feature extraction

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- PennyLane - quantum machine learning framework
- Streamlit - interactive web interface
- NumPy - numerical computation

8. How to Run

```
pip install -r requirements.txt
```

```
streamlit run app/app.py
```

Open <http://localhost:8501> and upload a product image

9. Test Results

Validated on three sample images:

- good_product.png -> GOOD (89% confidence)
- defect_scratch.png -> DEFECTIVE (69% confidence)
- defect_texture.png -> DEFECTIVE (89% confidence)

10. Evaluation Criteria Alignment

- Detection Accuracy (35%): Multi-metric classical + quantum hybrid fusion
- Innovation (20%): Full quantum vision pipeline with variational quantum classifier
- Efficiency (20%): Fast CPU-only inference, no GPU required
- Usability (15%): Drag-and-drop UI with heatmaps and downloadable reports
- Presentation (10%): Visual analysis panels, circuit diagram, metrics dashboard

11. Repository

GitHub: <https://github.com/Maheshwar422/qt-2.3-quantum-defect-detection>

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